

Giulio Arecco

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[LinkedIn ↗](#) | [GitHub ↗](#) | [Portfolio ↗](#)

Education

Politecnico di Torino, MS in Computer Engineering Sept 2023 – March 2026

- Specialization: Computer Graphics and Multimedia
- Final Mark: 110/110
- Master's Thesis: "Entertainment games for complex problem-solving" (Thesis PDF [↗](#))
- Relevant Coursework: Computer Graphics, Game Design, Virtual Reality, Computer Animation, System and Device Programming

Politecnico di Torino, BS in Computer Engineering Sept 2020 – March 2024

Selected Projects

Beyond The Line – Master's Thesis Serious Game [GitHub ↗](#) | [Itch.io ↗](#)

- Developed a text-based serious game featuring dynamic narrative branching, designed to evaluate players' Complex Problem Solving skills through resource management and systemic uncertainty.
- Tools Used: Unity, C#, Ink

The Plague in Milan – 3D Interactive Experience [GitHub ↗](#) | [Itch.io ↗](#)

- Engineered a physics-driven first-person controller, an event-driven modular progression system, and a dialogue system to interact with non-player characters.
- Tools Used: Unity, C#

Monte Carlo Path Tracer [GitHub ↗](#)

- Developed a dependency-free CPU path tracer featuring compile-time zero-cost polymorphism, explicit memory ownership, BVH acceleration, and custom À-Trous/Bilateral spatial denoisers.
- Tools Used: Zig

OS/161 Demand Paging [GitHub ↗](#)

- Implemented a demand-paging kernel subsystem for the OS/161 operating system featuring on-demand lazy ELF loading, a FIFO page replacement queue, disk-backed swap management, and software TLB handling.
- Tools Used: C, OS/161

Technologies & Skills

Programming Languages: C++, C#, C, Zig

Game Engines: Unreal Engine 5, Unity

Developer Tools: Git, JetBrains Rider, Visual Studio, WSL/Linux

Spoken Languages: English (C1), Italian (Native)